

CNTAC Fast Track Proposal - COVER PAGE

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- **Proposal information:**

Title: Census of optical counterparts of Einstein Probe Fast X-ray Transients

Abstract: Fast X-ray Transients (FXTs) manifest as short flashes of X-ray photons with durations ranging from minutes to hours; nevertheless, their origin is still a mystery. Proposed scenarios for FXTs involve: tidal disruption of a white dwarf by an intermediate-mass black hole; a binary neutron star merger, supernovae shock break-out; and off-axis or low-luminosity gamma-ray bursts. The successful launch of the X-ray mission Einstein Probe (EP) in January 2024 has enabled the discovery of many dozens of FXT candidates up to September 2024. Taking advantage of the quick, public detection announcements and relative localizations from EP, rapid optical follow up has already permitted counterpart identification in a fraction for the first time, disentangling their origin. Thus, we request imaging and spectroscopic for follow up 30 new FXTs detected by EP, using SPECULOOS+VST+PDTs facilities.

Telescope, instrument, time requested (both preferred and minimum acceptable time, preferred dates, Moon requirements, etc.):

* VLT Survey Telescope:

- 16 nights (4 nights min.)
- any dates (SM; ideally as 2h ToOs for flexibility, but full nights acceptable)
- any Moon phase

* Planetary Defense Tel. 1.3m:

- 156h (32h min.)
- any dates (QM; in 1-4h blocks)
- any Moon phase

* Planetary Defense Tel. 1.0m:

- 180h (36h min.)
- any dates (QM; in 1-4h blocks)
- any Moon phase

* SPECULOOS:

- 25 nights (5 nights min.)
- any dates (SM; full nights)
- any Moon phase

SCIENTIFIC AIM AND RATIONALE

Extragalactic Fast X-ray transients (FXTs): Prior to 2024, only ≈ 30 **extragalactic fast X-ray transients (FXTs, manifesting as few minutes to hours flashes)** had been discovered in archival Chandra, XMM-Newton and Swift data [1-7]. Extrapolation of their number density from these small-FOV observatories, however, means that FXTs are in fact ubiquitous, with $\sim 10^6$ occurring all-sky per year at the sensitivity of Chandra and XMM-Newton [2,7]; e.g., **$\sim 100\times$ more common than Gamma-Ray Bursts (GRBs), thus implying a huge knowledge gap among transients**. Host studies of these archival FXTs indicate that $>70\%$ lie at $z>0.5$ [7,13,15]. **Yet, their exact nature remains elusive** due to imprecise redshifts and almost no contemporaneous multi-wavelength constraints [e.g., 1–22].

Einstein Probe (EP) satellite: The recently launched EP satellite can potentially fill our knowledge gap, discovering >70 exciting and not-fully-understood new FXTs in the last ~ 10 months. The Wide-field X-ray Telescope (EP-WXT) is the first sensitive wide-field (3600 deg^2 field-of-view) monitor at soft X-ray energies (0.5–4 keV), designed to rapidly detect bright FXTs and alert the community. EP-WXT localizes FXTs to an uncertainty of $\sim 3'$, which can be improved to $\sim 10''$ by subsequent automatically triggered or scheduled observations with EP's Follow-up X-ray Telescope (EP-FXT) [38]. EP has reached a detection rate of $\sim 100 \text{ FXTs yr}^{-1}$ at high SNR, far exceeding expectations. Among announced FXTs, EP localizations have permitted the detection of 37 optical and NIR counterparts (including 16 persistent flaring stars and AGN). This is great, BUT a majority remain unclassified! **The goal of this program is to place uniform optical constraints on all observable FXTs.**

Several models have been put forth to explain FXTs (with different X-ray and optical evolution):

- **Binary Neutron Star Mergers (BNS):** Some models predict the formation of a magnetar that emits an isotropic, fast X-ray signal [27,39]. The potential link between FXTs and BNS should be a kilonova (KN), an isotropic, faint, fast-evolving transient with blue to red colors (see Fig. 1 top-left panels) [40,41]. If a magnetar is present, it may enhance the KN emission, especially in blue colors (Fig. 1). To confirm the KNe association, we expect to detect faint and quick optical emission during the first days, with redder emission later, and unique spectroscopic features. They are predicted to reach X-ray luminosities of $L_{\text{peak,X}}^{\text{BNS}} \lesssim 10^{46} \text{ erg s}^{-1}$ [25,26,27,28].

- **White dwarf (WD) + intermediate-massive black hole (IMBH) tidal disruption events (TDE):** WDs disrupted by IMBHs should produce FXTs [23,29,42]. According to simulations, the optical spectra and light curves of the disrupted material from WD-IMBH TDEs will show properties similar to thermonuclear transients [23], evolving from blue to redder colors on $\sim$ week timescales (see Fig. 1 top-right panels). To guarantee the TDE nature, spectroscopic confirmation showing thermonuclear features is needed. Their predicted X-ray luminosities span a wide range $L_{\text{peak,X}}^{\text{TDE}} \lesssim 10^{48} \text{ erg s}^{-1}$ [23,29].

- **Supernova (SN) Shock Breakouts (SBO):** some SN-SBO models predict FXTs, whereby the best-fit X-ray light curve parameters provide novel constraints on the progenitor star radius and outer density/CSM [30]. It is expected that the optical/NIR counterparts of SN-SBOs detected by EP-WXT will show SNe spectral features and light curves (type Ib/Ic) similar to SN 2008D (see Fig. 1 bottom-left panel) [31], further solidifying the one and only direct association to date between FXTs and SNe. It is expected that the SN counterpart may be visible some days after the X-ray trigger. Models predict they will reach an X-ray luminosity of $L_{\text{peak,X}}^{\text{SBO}} \lesssim 10^{44} \text{ erg s}^{-1}$ [30,31].

- **GRBs:** High- z , off-axis, and low-luminosity (LL) GRB afterglows can all produce FXTs, with multi-wavelength light curves and spectra showing a variety of behaviors. New FXTs such as EP 240315a [8], EP 240414a [14], EP 240806A [43], and EP 210801a [46] have demonstrated the importance of acquiring early optical/NIR imaging [9,16,45,47,48] (see Fig. 1, bottom panels), as well as spectroscopy to assess their redshifts [10,11,44], and ultimately formalize their potential associations with GRB afterglows. X-ray luminosities are predicted to be $L_{\text{peak,X}}^{\text{cGRB}} \sim 10^{44} - 10^{47}$ for cocoons/LLGRBs [30,31] and $L_{\text{peak,X}}^{\text{zGRB}} \geq 10^{47} \text{ erg s}^{-1}$ for high- z [49], respectively. 18/37 EP counterparts are associated with GRBs.

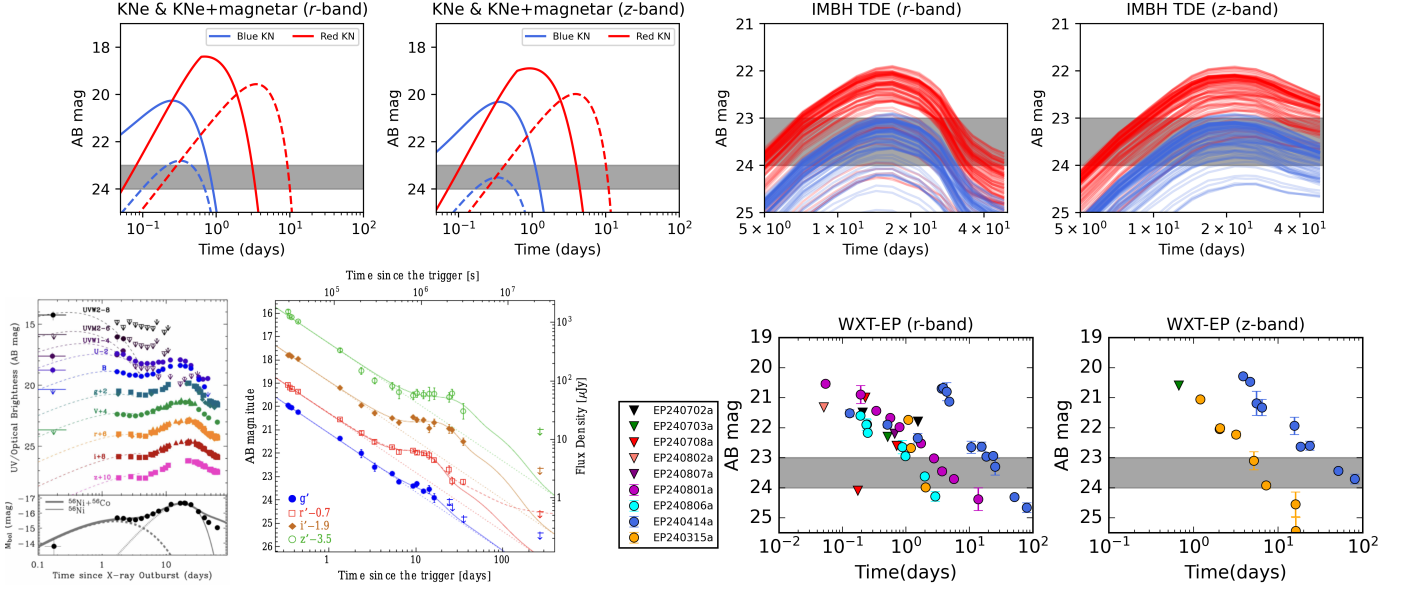


Figure 1: *Top-left panels:* Light curve evolution of blue and red KNe (*dashed lines*) and KNe+magnetar (*solid lines*) at $z=0.2$, using the numerical model proposed by [27,28], assuming different ejecta ($M_{\text{ej}}^{\text{blue}}=10^{-4}$, $M_{\text{ej}}^{\text{red}}=10^{-3} M_{\odot}$ and $\kappa^{\text{blue}}=0.5$, $\kappa^{\text{red}}=20 \text{ cm}^2 \text{ g}^{-1}$) and magnetar ($B_p \approx 10^{16} \text{ G}$) parameters. *Top-right panels:* Predicted light curves of a WD TDE from different viewing angles (using simulations from [23]). The different *red* and *blue* lines depict redshifts of $z=0.2$ and 0.3 , respectively. *Bottom panels from left to right:* Optical light curves of SBO+SN2008D, GRB101219B/SN2010ma, and a sample of detected (*circles*) or upper limits (*triangles*) optical counterparts of EP-discovered FXTs in the *r* and *z* bands. For the latter, the *grey* horizontal region is our proposed 3σ limiting magnitude. **Deep and uniform optical constraints from ~ 1 –60 days to depths of ~ 23 –24 ABmag have been sorely lacking thus far.**

Each progenitor channel can easily explain the timescales and X-ray spectra associated with a portion of the FXT population. **The rapid detection of multi-wavelength counterparts remains a critical first step to elucidate their nature**, enabling spectroscopy with larger telescopes and tracing the evolution of their emission mechanisms. **It is important to stress here that EP-WXT transients may potentially elucidate details on many high-profile and still poorly understood/constrained scientific areas**, such as: a) progenitors of gravitational waves and accelerated particles; b) the last days of massive stars; c) the source of lanthanides; d) the demographics of IMBHs; the physics of low-luminosity and off-axis GRBs (just to name a few).

Conditions to trigger observations: Extrapolating from known events, the **EP-WXT detection rate is $\sim 100 \text{ FXTs yr}^{-1}$** at a peak flux of $F_{X,\text{peak}} \gtrsim 5 \times 10^{-10} \text{ erg cm}^{-2} \text{ s}^{-1}$. This trigger flux is ≈ 10 times brighter than the 1 ks detection limit of EP-WXT. For events where the EP-FXT telescope provides an accurate position of ≈ 10 – $20''$, we will consider a trigger flux 2.5 times lower. The discovery of new EP transients are announced publicly via GCN circulars (typically $\lesssim 2$ hours, but as short as $\lesssim 4$ minutes), where initial EP-WXT localization regions are $\sim 3'$, reduced to $< 10''$ by subsequent triggered EP-FXT observations (or 3 – $6''$ if *Swift*-XRT observed). We will observe both extragalactic and Galactic announced targets. From Chilean sites, we can observe $\approx 70\%$ of the visible sky, **implying ≈ 35 targets during semester 2025A**. Observations will be cross-matched for identifications in ongoing time-domain surveys such as ZTF or ATLAS, similar to EP 240315a/AT2024eju [8,9], to avoid unnecessary observations.

Observing plan: To confirm KNe, we need observations on a few-day rest-frame timescale, while SNe or TDE origins require observations out to their rest-frame peak of $\approx 20\text{--}30$ days and collapsar/GRBs may be relatively bright or faint early on, depending on viewing angle, and show a broad-line SN Ibc peak around ≈ 30 days (see Fig. 1). Based on the 12 FXTs from EP-WXT with spectroscopic redshifts thus far, we assume SBOs, TDEs, and KNe might be seen out to $z \sim 0.5$, while GRBs out to $z \sim 5$. We thus aim for *griz* monitoring to ≈ 23.5 ABmag, with an approximate logarithmic cadence of 2/4/8/16/32/64 days, to place meaningful constraints on a large portion of the expected viable parameter space (non-detections will imply high-redshift channels). To ensure that our program has some robustness against local weather variations, we request to use available CNTAC fast track facilities SPECULOOS, VLT-Survey Telescope (VST), and 1.0m and 1.3m Planetary Defense Telescopes (PDTs) to implement a comprehensive ‘mid’-term optical follow-up program for all EP-discovered FXTs visible from the Southern Hemisphere. Given the number of expected targets, we will on average follow up 1 target intensely each week, after which longer term follow-up monitoring can be programmed in and adjusted a few days ahead of the schedule. The current CNTAC SPECULOOS schedule provides guaranteed observations every ~ 7 days, while the other 3 instruments are queue-scheduled and can be used to fill in the 7-day gaps. Our choice of *griz* photometry is aimed at isolating different theoretical FXT counterparts and exploring their color evolution as described above (e.g., KNe or IMBH TDEs might be initially bright in *u*; see Fig 1).

References: • [1] Jonker et al. 2013, ApJ, 779 14 • [2] Glennie et al. 2015 MNRAS, 450, 3765 • [3] Irwin et al. 2016, Nat, 538, 356 • [4] Bauer et al. 2017, MNRAS, 467, 4841 • [5] Xue et al. 2019, Nat, 568 • [6] Quirola-Vázquez et al. 2022, A&A, 663, 168 • [7] Quirola-Vázquez et al. 2023, A&A, 675, 44 • [8] Zhang et al., GCN 35931 • [9] Srivastav et al., GCN 35932 • [10] Saccardi et al., GCN 35936 • [11] Quirola-Vázquez et al., GCN, No. 35960 • [12] Gillanders, et al. 2024, arXiv:2404.10660 • [13] Levan, A.J., et al. 2024, arXiv:2404.16350 • [14] Liu, Y., Sun, H., Xu, D., et al. 2024, arXiv:2404.16425 • [15] Lian, T. Y., et al. 2024, GCN, No. 36091, 36091 • [16] Aryan, A., et al. 2024, GCN, No. 36094, 36094 • [17] Jonker, P. G., et al. 2024, GCN, No. 36110, 36110 • [18] Levan, A. J., et al. 2024, GCN, No. 36355, 36355 • [19] Bright, J., et al. 2024, GCN, No. 36362, 36362 • [20] Svinkin et al., GCN, No. 35972 • [21] DeLaunay, J. et al. GCN, No. 35971 • [22] Alp D. & Larsson J., 2020, ApJ, 896, 39 • [23] Kasen et al. 2017, Nature, 551(7678):80–84 • [24] MacLeod et al. 2016, ApJ, 819(1):3 • [25] Mazzali et al. 2008, Science, 321(5893):1185. • [26] Z. G. Dai et al. 2006, Science, 311(5764):1127–1129. • [27] B. D. Metzger, E. Quataert, et al. 2008, MNRAS, 385(3):1455–1460. • [28] Y. W. Yu et al. 2013, ApJ, 776(2):L40. • [29] H. Sun et al. 2017, ApJ, 835(1):7 • [30] K. Maguire et al. 2020, Space Sci. Rev., 216(3):39 • [31] Eli Waxman and Boaz Katz, 2017, In Handbook of Supernovae, page 967. • [32] A. M. Soderberg et al. Nature, 454(7201):246. • [33] Ehud Nakar and Tsvi Piran, ApJ, 834(1):28. • [34] Fabio De Colle et al. 2018, MNRAS, 478(4):4553–4564. • [35] Gaia Collaboration 2021, A&A, 649:A1. • [36] Andrea De Luca et al. 2020, A&A, 634:L13. • [37] David García-Alvarez et al. 2008, ApJ, 679(2):1509–1521. • [38] Phillips et al. 2020, A&A, 637, A38. • [39] W. Yuan et al. arXiv preprint arXiv:1506.07735, 2015. • [40] Zhang 2013, ApJ, 763(1):L22. • [41] V. A. Villar etl. 2017, ApJ, 851(1):L21. • [42] S. J. Smartt et al. 2017, Nature, 551(7678):75–79. • [43] R. Saxton et al. 2021, Space Sci. Rev., 217(1):18. • [44] Q. Y. Wu et al. GCN, No. 37063. • [45] Quirola-Vázquez et al. GCN, No. 37087 • [46] Pérez-Fournon et al. GCN, No. 37080 • [47] Zhou et al. GCN, No. 36997 • [48] Fu et al. GCN, No. 36998 • [49] Quirola-Vázquez et al. GCN, No. 37013 • [50] Evans et al. 2010, A&A, 519, A102

TECHNICAL DESCRIPTION

Due to the unknown nature of FXTs, our main goal is to characterize (via optical imaging) ~ 30 FXTs detected by EP mission during semester 2025A at declination $\delta < 20^\circ$. We expect to follow up EP-discovered FXTs using SPECULOOS+VST+PDTs facilities. As stated previously, we aim for an approximate logarithmic cadence of 2/4/8/16/32/64 days for each FXT. We estimate 1 hr exposure per filter to reach $r \approx 23.5$ AB mag for the 1m-class telescopes, while and 0.5 h for VST. We require $2\times$ exposure in g in bright time (amounting to 25% more time required in g on average, and hence 6% more time overall). We adopt 8–9h per night as the average observable time. We also assume that 20% of the fixed SPECULOOS nights will be lost to weather.

Adopting a scheme of 2 epochs with VST and 4 epochs with 1m telescopes, to carry out the above program we estimate that we need 30 objs x 4 filters x 2 epochs x 0.53h $\approx$ 130h on VST and 30 objs x 4 filters x 4 epochs x 1.06h $\approx$ 510h on 1m-class imagers, respectively. Thus, we request 16 nights (136hr) on VST, 25 nights (212h*0.8=170h) on SPECULOOS, 180h on 1.0m and 156h 1.3m PDTs.

If less time is available, we will reduce the number of FXTs or number of epochs for follow-up; the minimum time for this proposal to make any realistic impact would be $\approx 20\%$, with VST being the most important component, followed by Planetary Defense telescopes, and then SPECULOOS (due to more rigid and heterogeneous scheduling and policies).

CURRENT STATUS OF THE PROJECT

Collaborators of this highly productive project have been awarded with telescope time to follow-up potential counterparts of FXTs in different energy bands: *Swift-XRT telescope*: 12 ksec; *MeerKAT*: 8 hours; *ESO-VLT*: 6.9 hours MUSE for follow-up; *GTC*: 34 hours HiPERCAM; *GTC*: 10 hours OSIRIS and EMIR; *NOT*: 1.8 nights ALFOSC; *XMM-Newton*: 50 ksec.

The collaboration is applying to ground and space facilities for additional time allocation to ensure coverage across a range of observable windows (declination, time zones, weather). It is important to stress that the majority of the awarded optical and infrared facilities above just cover the northern hemisphere (except VLT-MUSE). Thus, the relevance of this proposal is because it will cover the Southern hemisphere, solving the sky localization gap of FXTs. Furthermore, due to the quick announcements of FXT detections (typically $\lesssim 2$ hours) and highly accurate localization done ($< 10''$) by EP's instruments, the requested time of this proposal will permit to identify potential optical counterparts and acquire spectra during the first few days after the X-ray trigger. Indeed, during the last months, EP has detected a couple of FXTs, which have been observed by our group (imaging and spectroscopy) unmasking their true nature (papers in preparation). The related experience of this collaboration started some years ago. A selection of relevant refereed publications about FXTs of the last years from this team is summarized as follows (this includes several papers by former PUC PhD student Quirola-Vásquez, highlighting the strong Chilean involvement and productivity):

(1) van Dalen, J. et al. (2024), "The Einstein Probe transient EP240414a: Linking FXTs, GRBs and Luminous Fast Blue Optical Transients", A&A, in press. (2) Inkenhaag, A. et al. (2024), "Redshifts of candidate host galaxies of four FXTs using VLT/MUSE", A&A 689, A343. (3) Wichern, A. et al. (2024), "Investigating the off-axis GRB afterglow scenario for extragalactic FXTs", A&A, accepted. (4) Liu, Y., et al. (2024), "Soft X-ray prompt emission from a high-redshift GRB EP240315a", Nature, submitted. (5) Levan, A., et al. (2024), "The FXT EP240315a: a $z \sim 5$ GRB in a Lyman continuum leaking galaxy", Nature, submitted. (6) Wang, L., et al., (2024), "Newly formed dust within the circumstellar environment of SN Ia-CSM 2018evt", Nature Astronomy, 8, 504. (7) **Quirola-Vásquez, J., et al.** (2024), "Probing a Magnetar Origin for the population of Extragalactic FXTs detected by *Chandra*", A&A 683, A243, March, 2024. (8) Eappachen, D., et al. (2024), "XMM-Newton-discovered FXTs: Host galaxies and limits on contemporaneous detections of optical counterparts", MNRAS, Volume 527, Issue 4, February 2024. (9) **Quirola-Vásquez, J., et al.** (2022), "Extragalactic FXT candidates discovered by Chandra (2004-2022)" A&A, 675, 44, July 2023. (10) Eappachen, D., et al. (2022), "The FXT XRT 210423 and its Host Galaxy", ApJ, 948, 1, may, 2023. (11) **Quirola-Vásquez, J., et al.** (2022), "Extragalactic FXTs discovered by Chandra (2000-2014)" A&A, 663, A168, July 2022. (12) Eappachen, D., et al. (2022), "Probing for the host galaxies of the FXT XRT 000519 and XRT 110103", MNRAS, 514, 1, July 2022. (13) Yang, G., et al. (2019), "Searching for extragalactic FXTs in Chandra surveys," MNRAS, 487, 4721-4736. (14) Sun, H., et al. (2019), "A unified binary neutron star merger magnetar model for the Chandra X-ray transients CDF-S XT1 and XT2," ApJ, 886, 2, 129. (15) Xue, Y. Q., et al. (2019), "A magnetar-powered X-ray transient as the aftermath of a binary neutron-star merger," Nature, 568, 198-201. (16) **Bauer, F. E., et al.** (2017), "A new, faint population of X-ray transients," MNRAS, 467, 4841-4857. (17) Glennie, A., et al. (2015), "Two FXTs in archival Chandra data," MNRAS, 450(4), 3765-3770. (18) Jonker, P. G., et al. (2015), "Discovery of a New Kind of Explosive X-Ray Transient near M86," ApJ, 779, 14.

STUDENT THESIS

The proposed follow-up forms an important component of one PhD project (graduate student in a Chilean institution). The PhD thesis focuses on the study of the origin of FXTs and starts this semester. Data/detections of optical counterparts of FXTs play an important role because other energy bands give unique information to explore potential origin(s). From this dataset, the student will lead at least one optical follow-up publication on a key counterpart detection, be a co-I on 5-10 additional publications, and lead a publication on the overall distribution of counterpart detections/limits for a large sample of FXTs.